

TIM CALKINS

Amgen: Launching Repatha

Sheila Bush, senior vice president and head of U.S. commercial operations at biotechnology firm Amgen, looked at Alex Shamoun, business team leader for Amgen's new cholesterol-lowering drug Repatha (evolocumab), and smiled. It was March 7, 2015, a warm spring day in California. Bush had just finished reviewing Shamoun's marketing plan for Repatha. Like most industry experts, Bush and Shamoun anticipated that the FDA would approve the molecule in August 2015.

"This marketing plan is really coming together," observed Bush. "We all believe Repatha will be the next blockbuster for Amgen, especially after our long-term trials are completed in 2019. It has enormous potential." Shamoun nodded, pleased that his plan had been well received.

"There is just one problem, Alex," noted Bush. "We need to develop a solid revenue forecast. We can't just propose a marketing plan without some solid objectives. Let's meet again next week; I want to review your U.S. revenue forecast for Repatha for 2018."

With that, Bush stood up, looked at her phone, and headed to her next meeting.

Shamoun quickly realized this would be no easy task. To assess the potential 2018 revenue for a brand new drug, he and his team would have to determine the target patient population, establish an estimated price, and generate market share estimates, as several competitors would also soon launch drugs in this category.

Amgen

Amgen was one of the world's leading biotechnology companies, with more than 20,000 employees and \$18 billion in revenue. The company was founded in 1980 under the name Applied Molecular Genetics and assumed the name Amgen in 1983.

Amgen was doing well in 2015. It was highly profitable: in 2014, it had net income of \$5.2 billion on revenues of \$20.1 billion (see **Exhibit 1**). Over the prior two years, Amgen's revenue had grown at a CAGR of 7.8%. Profitability grew at an even faster rate, up 10.0%. Amgen's stock did exceptionally well, climbing from \$85 per share at the start of 2013 to \$160 at the start of 2015. Market capitalization was over \$120 billion.

Amgen had six drugs with over \$1 billion in revenues. The most important were Aranesp and Epogen, for anemia; Enbrel, for arthritis; and Neulasta, for febrile neutropenia, a condition associated with chemotherapy (see **Exhibit 2**). These key products accounted for the vast majority of Amgen's revenues. Amgen also had several products experiencing rapid revenue growth, including Prolia and Xgeva. The U.S. market was particularly important for Amgen, as it made up over 80% of all of Amgen's revenue.

Given its strong financial performance, Amgen executives were highly focused on company growth. Amgen had several co-development deals in place with other large pharmaceutical firms such as GlaxoSmithKline, Takeda Pharmaceuticals, and AstraZeneca, and had purchased several other biotechnology firms in 2012 with hopes of finding strong new drug candidates and expanding its presence in emerging markets. Amgen was particularly focused on new drugs in oncology, neurology, immunology, and cardiovascular spaces.

Amgen's strong financial results also gave it the resources it needed to invest in R&D and commercialization. The company had several molecules of its own in development. One of the most important and promising of these molecules was Repatha, for high cholesterol.

Heart Disease and High Cholesterol

Cardiovascular disease was the leading cause of death in the United States, responsible for 2,200 deaths a day on average in 2013, or 800,000 each year. In fact, one out of every three deaths in the United States was caused by heart disease. In addition, heart disease caused heart attacks and strokes, which often left people in poorer health following the incident. The costs of heart disease in the United States in 2011, including healthcare expenditures and lost productivity, totaled over \$300 billion.¹ According to the American Heart Association, 85.6 million Americans were living with cardiovascular disease.²

High cholesterol (dyslipidemia) is one of the major controllable risk factors for coronary heart disease, heart attack, and stroke. As cholesterol rises, so does the risk of coronary heart disease. Cholesterol is a waxy substance both created by the human body (mainly in the liver) and ingested through food. Though high cholesterol often comes with dangerous side effects, the body does need a certain amount of cholesterol to circulate through the blood in order to perform regular bodily functions. The body produces enough cholesterol on its own to perform those functions. Foods high in saturated and trans fats, such as red meat and full-fat dairy products, can raise cholesterol to unhealthy levels.

Cholesterol comes primarily in two forms: low-density lipoprotein (LDL), commonly referred to as "bad cholesterol," and high-density lipoprotein (HDL), commonly called "good cholesterol." Correspondingly, health problems from high cholesterol are associated with high levels of LDL, whereas high levels of HDL are presumed to be positive. LDL cholesterol is considered harmful because it creates plaque, which is the buildup of fatty deposits in the arteries. With excess plaque,

arteries become less flexible and the body is more apt to have reduced flow of blood and oxygen. This condition, known as atherosclerosis, leads to blood clots, which are a main factor in heart attacks and strokes.

Typically, high cholesterol is diagnosed through a routine blood test that measures total cholesterol, HDL, LDL, and triglycerides. In the United States, cholesterol levels are measured in milligrams per deciliter (mg/dL) of blood. A desirable level of total cholesterol is considered to be less than 200 mg/dL, whereas more than 240 mg/dL is a significant concern. Ideal levels of LDL vary based on a person's risk factors, such as familial history of heart disease. Specifically, the ideal level of LDL is between 100 mg/dL and 129 mg/dL, but is less than 100 mg/dL for those at higher risk of heart disease. LDL cholesterol totals of 160 mg/dL or greater are considered to be high and could require intervention.³ Many experts expected that target LDL cholesterol levels would become even lower in the near future.

Approximately 75 million American adults (one in every three) had high LDL in 2013. Of this group, 30.9 million adults had LDL of over 240 mg/dL. Although 34 million adults had been treated for high cholesterol, only 23 million had their cholesterol under control.⁴ This left 52 million people with high LDL levels. As a result, high cholesterol was a very prominent issue in the United States, even with medication widely available.

Familial Hypercholesterolemia

Although most people can reduce their levels of LDL cholesterol by eating properly and avoiding foods with high saturated fats, some inherit genetic conditions that make high LDL more difficult to control. One of the more serious genetic conditions resulting in high cholesterol is called familial hypercholesterolemia (FH), and it often leads to premature cardiovascular disease. In FH patients, genetic mutations make the liver incapable of metabolizing excess LDL. As a result, individuals with FH are unable to recycle the natural cholesterol their bodies produce, resulting in higher cholesterol levels overall. Individuals with FH have a rate of heart disease twenty times higher than average, and can have LDL cholesterol counts of 400 or even 500 mg/dL.⁵

There are two forms of FH: heterozygous FH (HeFH), a genetic mutation that the individual inherited from one parent, and homozygous FH (HoFH), a mutation inherited from both parents. HeFH is found in about one in every 200 to 500 people worldwide, while HoFH is rarer and found in about one in every 160,000 to 1,000,000 people. Though HoFH is less common, it is extremely dangerous and severe in its consequences; in HoFH patients, atherosclerosis can start even before birth and progresses rapidly afterward. In fact, if left untreated, HoFH can lead to heart attacks as early as the teenage years. In 2013, there were about 625,000 individuals in the United States with some form of FH.⁶ Only about 1,000 of these individuals had HoFH; the rest had HeFH.⁷

Though treatable, FH was extremely hard to diagnose and over 90% of FH cases were not properly identified. The basic signs of FH were patterns of early heart attacks in the family (before age 55 for men and age 65 for women). All FH patients required cholesterol-lowering medications, and many required more extreme interventions such as LDL aphaeresis, a procedure to reduce cholesterol within the blood. Some patients required weekly aphaeresis treatments, with each session taking up to three hours. Annual costs for the therapy could exceed \$100,000.

Pharmaceutical companies had been searching for solutions to high cholesterol and FH for years, with mixed success. However, more recently, scientists discovered a major class of drugs called statins that helped control cholesterol with manageable side effects. These drugs were generally quite effective for individuals with high cholesterol, though they were not particularly effective for individuals with FH.

Statins

In August 1973, two Japanese scientists named Akira Endo and Masao Kuroda discovered a compound that inhibited the enzyme HMG-CoA reductase, a protein that aided in the production of LDL. Inhibiting the HMG-CoA reductase enzyme slowed the pace of LDL production, allowing a greater absorption of LDL by the liver and ultimately lower amounts of LDL in the bloodstream.

This discovery led to the production of statins, starting with Endo and Kuroda's mevastatin in 1976. While that product never entered the market due to its severe side effects, it led to the 1987 launch of the first statin in the United States: Mevacor, manufactured by Merck. Trial data for Mevacor was strong, showing that it reduced LDL cholesterol levels by an average of 25%.⁸ Merck's marketing efforts focused on its sales force, which developed close relationships with physicians and encouraged them to prescribe Mevacor, and on direct-to-consumer (DTC) advertising, which helped educate patients on the effects of high cholesterol and how Mevacor could help. Merck's first statin competition did not come until 1991, when Bristol-Myers Squibb launched Pravachol, which had a similar effect as Mevacor on LDL levels. Mevacor continued to be the market share leader in the growing statin market even after the launch of Pravachol.

Though sales for Mevacor reached nearly \$1 billion by 1991, many felt the statin market had not yet matured, as trials on Mevacor and Pravachol proved only that they could lower LDL levels, not prevent a heart attack or other serious cardiac events.⁹ If a statin's trials could prove the latter, more physicians would be persuaded to prescribe it. Merck launched Zocor in 1992 with the data required to expand the statin market. Not only did Zocor lower LDL cholesterol levels by 35%, but trials also showed that patients on Zocor were 42% less likely to die from a heart attack than patients who took a placebo.¹⁰ As a result, the market greatly expanded by 1995, with Mevacor and Zocor the clear market leaders.

The market grew even further in 1998 with Pfizer's launch of Lipitor, the first statin to have an indication for HoFH. Lipitor succeeded for a variety of reasons. First, Pfizer conducted trials not just against a placebo, but also against other statins, including Zocor and Pravachol, showing its superiority in head-to-head competitions at reducing cholesterol. In fact, at every dosing level, Lipitor outperformed all other statins on the market in LDL reduction.¹¹ These head-to-head results were extremely persuasive, and Lipitor boomed even though Pfizer did not complete cardiac outcome trials (as Merck had for Zocor) prior to releasing it. Second, Pfizer launched extremely aggressive marketing that stressed Lipitor's potency and price, which was about 9% below that of Zocor.¹² Even though Lipitor was the fifth drug to enter the statin market, it became the market leader by 1999. It helped grow the statin market tremendously, making statins the most profitable class of drugs in the world, with 118 million prescriptions and \$10.7 billion in U.S. sales by 2002.

Even with intense competition, Lipitor remained the market share leader in statins throughout its life on patent, achieving more than \$13 billion in global sales in 2006. Pfizer spent over \$300 million in marketing annually in the United States, reaching a peak of \$660 million in 2010,

with nearly half of the marketing efforts spent on DTC campaigns.¹³ Even the 2003 launch of AstraZeneca's statin, Crestor, which had superior efficacy in head-to-head LDL-lowering trials against Lipitor and similar marketing expenditures, could not remove Lipitor from its perch as the leader in the statin market.

Statin market revenues began to decline as products lost patent protection. When a pharmaceutical product loses patent, other drugs using the same formula are able to enter the market at a much lower price than the brand-name product. As a result, competition increases while revenue decreases, as brand manufacturers lower prices of their drugs to compete with generic entrants. In the case of statins, Zocor went off patent in 2006 and Lipitor's patent expired in 2011. As expected, revenue fell dramatically. Specifically, Zocor's U.S. revenue fell from \$3.1 billion in 2005, the last full year on patent, to under \$1 billion in 2007. After achieving nearly \$5.3 billion in U.S. sales in 2010, the last full year it was on patent, Lipitor's U.S. sales dropped to \$1.6 billion in 2012.¹⁴ As of 2015, the only statin whose patent had yet to expire in the United States was Crestor, though its patent would expire in July 2016. However, in the face of generic statins, Crestor saw its annual U.S. revenue fall from a high of nearly \$8 billion annually down to \$5.3 billion in 2013.¹⁵

In 2015, statins were a broadly used, well-established class of drugs. In the United States alone, 25 million people were prescribed a statin. The drugs were effective, reducing cholesterol levels by 30% to 50%, and generally safe. Statins were also cheap, with a generic version of Lipitor selling for about 40 cents per dose. Crestor, the only leading statin still protected by a patent, cost about \$6.57 per dose.

Still, there were unmet needs. First, many people who could benefit from a statin were not yet taking one. Although 75 million adults had high cholesterol, only 34 million had been treated. Second, compliance was a problem. People with high cholesterol were supposed to take a statin every day for the remainder of their lives, but many did not. About 50% of patients discontinued statin use within one year. Among adults, only 25.4% of patients were still compliant at the end of two years.¹⁶ Third, statins did not work for everyone. Some people, such as those with FH, did not respond at all, or did not respond enough to bring their cholesterol under control. Others could not take statins due to their side effects; some experts estimated almost 30% of patients on statins experienced muscle pain.¹⁷ Finally, heart disease continued to be a problem. Despite lower cholesterol levels, heart disease remained one of the leading causes of death.

Key Players

Patients usually learned of their high cholesterol diagnosis from visits to primary care physicians through routine blood tests. Typically, physicians would initially recommend dietary and exercise changes. If the patient still had high LDL several weeks after the initial diagnosis, the physician would generally prescribe a statin at a low dose (such as 10 mg). Every few weeks, patients saw their physician, who monitored their progress.

Physicians were generally comfortable using statins; the drugs had been on the market for many years. However, some doctors were wary of side effects and risks, especially at higher doses, and many of them left patients at a low 10 mg dose even if a patient was not fully to goal.¹⁸

Insurance companies were very sensitive to pricing and costs; they were reluctant to pay for expensive medications without compelling data. Most payers relied on pharmacy benefits

managers (PBMs), such as Express Scripts, to make coverage decisions and interact with patients and physicians. Most PBMs required extensive documentation, such as a detailed patient history and test results, before approving an expensive medication. This administrative burden often fell on the treating physicians and their staff. Payers generally covered the cost of statins, however, especially drugs that had lost patent protection and were inexpensive.

PCSK9 Inhibitors

Realizing there was still a large group of patients with unmet needs for high cholesterol, pharmaceutical companies began searching for a new remedy for LDL cholesterol issues in the early 2000s. Scientists found a potential solution in drugs that attack the proprotein convertase subtilisin/kexin type 9 enzyme, more commonly referred to as PCSK9. PCSK9 is an enzyme that interferes with the body's ability to remove LDL cholesterol. Specifically, PCSK9 targets LDL receptors for degradation, severely hampering the body's ability to remove LDL from the blood and causing the body to overproduce LDL cholesterol. Consequently, efforts to inhibit PCSK9 would reduce LDL cholesterol production. With promising initial concept testing and trials by scientists, pharmaceutical companies began to investigate potential PCSK9 inhibitor therapies in the early 2010s.

Unlike statins, PCSK9 inhibitors were injectable drugs, taken either once every two weeks or once a month.¹⁹ Patients would do the subcutaneous injection at home, injecting themselves in the thigh, upper arm, or abdomen, similar to insulin shots for those with diabetes. Some of the new products used a pre-filled syringe, while others had a spring-based auto injector or an automated mini-doser.

PCSK9 inhibitors were a type of biologic drug. Biologics were different from most drugs because they were manufactured with extracted biological sources. Examples of biologic products included vaccinations, gene therapies, and cancer antibodies such as Avastin, Herceptin, and Rituxan. Given their additional complexity to manufacture, biologics like PCSK9 generally cost significantly more than traditional small molecule drugs.

As of early 2015, no PCSK9 drugs had yet reached the market, though several were in development and expected to launch over the next few years. Industry experts believed that PCSK9 drugs would become a significant new class of drugs in the pharmaceutical industry, with the capability to help those for whom statins did not work. It was expected that initially, PCSK9s would be taken as a second-line therapy to statins, or taken with a statin to optimize treatment.

With PCSK9 inhibitor therapies showing early promise in drug development, several pharmaceutical firms began developing treatment options, which reached late stage trials by the end of 2014. The three firms who led the initial development of PCSK9 inhibitors were Amgen, with Repatha (evolocumab); Sanofi (partnered with Regeneron), with Praluent (alirocumab); and Pfizer, with bococizumab.

Amgen: Repatha (evolocumab)

Amgen's PCSK9 inhibitor, Repatha (evolocumab), first became known to the public in early 2012, when Amgen released its preliminary data. The drug showed promise almost immediately,

and Amgen executives referred to Repatha as “the biggest single thing we have in the pipeline” as early as November 2012.²⁰ Early-stage trials in 2012 showed that patients who were intolerant of statins experienced a 51% reduction in LDL after 12 weeks of therapy with Repatha.²¹

Such promising results and hype within Amgen led to swift commencement of a flurry of Phase III trials, which began in 2013 and continued through mid-2014. All had positive results. The trials showed that on its own, Repatha reduced LDL cholesterol by 57% when compared to a placebo, and up to 75% when combined with statin therapy.²² Furthermore, in a trial consisting exclusively of HoFH patients, Repatha showed a significant 31% reduction in LDL levels as compared to a placebo, giving hope that PCSK9 inhibitors would achieve cholesterol improvements for FH patients where statins were not as successful.²³ In addition, though early trials had indicated some potential neurocognitive impairments as a side effect, Amgen’s Phase III results indicated that patients were essentially free of major side effects. (See **Exhibit 3** for a summary of select trial results for PCSK9 inhibitors.) In fact, the most common adverse effects of Repatha were shown to be headache, diarrhea, nausea, and urinary tract infection, giving it an overall strong safety profile.²⁴ Results were similar for both biweekly and monthly administration of the drug, so analysts expected Amgen to focus on a monthly dosage.

As a result of these trials’ successes, Amgen submitted a biologics license application to the FDA on August 28, 2014, the first company to submit a license for a PCSK9 inhibitor. In the application, Amgen provided data for over 6,800 patients across ten trials.²⁵ Though Amgen’s trial data showed strong results that Repatha reduced LDL cholesterol, it did not yet show that Repatha would reduce cardiac events. Amgen had a number of long-term trials in place where such data would be measured, but the full data was not expected to be available until 2019. Results from an initial read-out of a long-term trial, released in 2015, indicated that Repatha indeed reduced the risk of cardiovascular events such as heart attack or stroke. In the initial read-out, 2.1% of patients on statins alone experienced a cardiovascular event. This percentage fell to just 0.95% for patients on both Repatha and a statin. Still, to claim that Repatha reduced cardiovascular events, Amgen would have to wait for broader, larger additional trials with positive results.²⁶ Specifically, if Amgen’s long-term trials showed a significant reduction in deaths, heart attacks, and strokes, it could file the data with the FDA to expand Repatha’s indication to include reducing risk of cardiac events. This indication could lead to significantly wider use of Repatha. However, given the timing of the long-term trials, most industry experts did not expect a supplemental filing until 2019.

The FDA’s typical review time from application to submission was about one year. Correspondingly, individuals within Amgen and in the broader medical community expected that Repatha would be available for prescription in August 2015.²⁷

Because it was the first to file its application to the FDA, Amgen was expected to lead the PCSK9 inhibitor race. Most experts believed being the first mover in the market would result in Amgen gaining the highest initial and sustained PCSK9 inhibitor market share. Though Amgen was intent on developing a cardiovascular pipeline, the company historically had not been particularly active in that space, so being a first mover was particularly important. In fact, Repatha and chronic heart failure drug Corlanor (ivabradine) were slated to be the first two major drugs for Amgen in the cardiovascular category. However, Amgen had significant competition in the race to bring a PCSK9 inhibitor to the market, which could greatly alter any revenue potential.

Sanofi: Praluent (alirocumab)

While Amgen was developing Repatha, Sanofi and Regeneron partnered to create another PCSK9 inhibitor, Praluent (alirocumab). The drug was initially developed and became known to the pharmaceutical and medical communities around the same time as Repatha, and underwent a similar trial regimen to Repatha. By July 2012, Sanofi was planning for Phase III trials, which would include over 22,000 patients in ten clinical trials. This drug was particularly meaningful for Sanofi for several reasons. First, the company was struggling in the cardiovascular market as its other medications were dealing with significant generic competition. Second, Sanofi's revenue and net income had been rapidly declining in recent years. Finally, the company had a very public shake-up when the board ousted the CEO in October 2014 due to disagreements over company strategy and direction.²⁸

Sanofi titled its Phase III program for Praluent the ODYSSEY program. Like Amgen's program, it was quite successful. Specifically, patients with high LDL at risk for cardiovascular disease experienced a 54.1% drop in LDL cholesterol relative to a placebo after 24 weeks of doses given every two weeks.²⁹ Another trial indicated that for individuals who were statin intolerant, Praluent was over ten times more effective than Merck's Zetia, the only non-statin available in the market (though not a PCSK9 inhibitor).³⁰ These results were similar to Repatha's. Later trials showed that patients with hypercholesterolemia who used Praluent could meet their cholesterol goals equally well through once-a-month dosing as with dosing once every two weeks, meaning that, like Amgen, Sanofi and Regeneron would likely aim for once-a-month dosing upon launch. Furthermore, additional Phase III results showed that Praluent was just as effective taken on its own as with a statin.³¹ Press reports were extremely positive, with several articles indicating that Praluent could reduce a patient's cholesterol level "to that of a baby."³² Praluent generated these results with essentially the same safety profile as Repatha, with limited adverse side effects and no evidence of neurocognitive impairments in later trials.

Like Amgen with Repatha, Sanofi was also gathering long-term data for Praluent. Specifically, Sanofi was able to link Praluent to reduced cardiac events in a preliminary long-term trial read-out, the results of which were released in 2014, before submitting an application to the FDA. The trial showed that Praluent approximately halved the number of heart attacks and strokes in HeFH and high-risk cardiovascular patients: 1.4% of Praluent-treated patients had a major cardiovascular event as compared to 3% for a placebo.³³ As the Phase III data yielded positive results, Sanofi and Regeneron were preparing to file a biologics license application as early as the end of 2014. This timeline would typically put Praluent several months behind Repatha in the race to be first to the market, but Sanofi and Regeneron purchased a priority review voucher from BioMarin Pharmaceutical for \$67.5 million in July 2014. The FDA occasionally gave out these vouchers in return for development of drugs for neglected diseases. With the voucher, Praluent would receive an expedited review process from the FDA, lasting just six months rather than the traditional ten to twelve.³⁴

In response to Sanofi's purchase of the voucher, Amgen sued Sanofi and Regeneron for patent infringement, citing that the companies used Amgen's PCSK9 intellectual property in creating Praluent.³⁵ However, as of early 2015, it seemed that the lawsuit would not delay Praluent's approval process, and that Amgen would receive small royalties on the revenues Praluent ultimately generated instead.

Sanofi and Regeneron filed a biologics license application for Praluent in late 2014, and the application was accepted on January 24, 2015.³⁶ Like Amgen, Sanofi and Regeneron did not have enough long-term data to include an indication for reduced risk of cardiac events (and general dyslipidemia) in this initial application. Instead, it was expected that Sanofi and Regeneron would submit further long-term trial results in 2019 and would receive expanded indications from the FDA that year.

Because of the priority waiver, Praluent's initial target action date became July 24, 2015, about a month before Repatha's, giving it an apparent first-mover advantage. However, it was possible that the FDA would not abide by its deadline, or that it would approve both drugs at the same time considering they had such similar target activation dates, biologic makeups, and patient targets.³⁷

Sanofi launched an early marketing plan for Praluent called "Cholesterol Counts," which featured an interactive website that allowed participants to learn about cholesterol levels, the consequences of high cholesterol levels, and treatment options. (See **Exhibit 4** for pictures of the campaign.) Website visitors were encouraged to participate in a "cholesterol knowledge" poll, with the results presented later in 2015, presumably in conjunction with the launch of Praluent.³⁸

Pfizer: Bococizumab

Though Pfizer began development on its PCSK9 inhibitor, called bococizumab, around the same time as Amgen and Sanofi, it was slower to create and complete trials, putting it approximately a year behind Amgen and Sanofi in the submission of its biologics license application to the FDA. Whereas Repatha and Praluent were expected to be on the market in July or August 2015, bococizumab was expected sometime in 2017.³⁹

As with Repatha and Praluent, initial results for bococizumab were impressive. Specifically, in Phase II trials, bococizumab reduced LDL cholesterol by 52% on patients also taking statins.⁴⁰ Bococizumab's Phase III trial results were expected to be complete in 2015 or 2016. Though initially Pfizer had planned for the long-term cardiovascular risk trials to be completed by 2017, a year ahead of most of Amgen's and Sanofi's long-term results, the company expanded the trials' size in 2014. Consequently, the results would be delayed until 2018 or 2019, with long-term cardiac outcomes data available at about the same time as those for Repatha and Praluent.⁴¹

Though bococizumab would not be the first PCSK9 inhibitor to the market, Pfizer was positioning it instead to be a "best in class" therapy that would steal significant share. To do so, Pfizer partnered with Halozyme Therapeutics, a biotechnology company that developed enzymes that carry drugs into the bloodstream. Halozyme's top product, called Enhance, broke down acids to help the efficacy of injection therapies. When Enhance was used with bococizumab, the number of necessary injections for therapy could be reduced to even less frequently than once per month.⁴² That would appeal not only to patients wary of injection therapies, but also to insurance companies concerned over potential costs of PCSK9 inhibitor treatment, as costs could potentially be reduced through fewer treatments.

Pfizer was looking into additional options for entry into the PCSK9 inhibitor market beyond bococizumab. Specifically, the company was working on a small molecule that also attacked the PCSK9 enzyme, with human trials set to begin in late 2015.⁴³ Many experts referred to a pill that attacked the PCSK9 enzyme as a "holy grail" in cholesterol lowering, which could ultimately trump

any injected therapy. Pfizer was also planning Phase I trials in 2016 for a PCSK9 vaccination that would be taken once per year.⁴⁴ These therapies, although promising, would not enter the market for several years, even if they were successful in trials.

Pfizer hoped bococizumab and its other PCSK9 inhibitor options would help it raise annual revenues, as the company had not achieved the same revenue level since Lipitor moved off-patent. Total annual company revenues fell from \$65.1 billion in 2010 (Lipitor's last year on patent) to only \$51.5 billion in 2013.⁴⁵ Pfizer needed a blockbuster and would potentially be willing to invest in heavy marketing for a strong, efficacious product like bococizumab or its small molecule, as it did with Lipitor, to win this fiercely competitive battle. Furthermore, Pfizer's extensive prior experience in the dyslipidemia and cardiovascular space could give it an edge over its less experienced competitors in the PCSK9 market once its products launched.

Other Competitors

Though Amgen, Sanofi, and Pfizer were the farthest along in developing PCSK9 inhibitor treatments, other companies were also in the process of conducting trials to bring additional PCSK9 inhibitors into the market. Roche, Bristol-Myers Squibb, Eli Lilly, and Alnylam all had drugs in development that could potentially challenge any of the first movers in the PCSK9 space.⁴⁶

In addition, another class of drugs called CETP inhibitors was also undergoing Phase III trials in early 2015. CETP inhibitors helped moderate cholesterol levels, moving triglycerides between LDL and HDL, and enhancing HDL counts while lowering LDL counts. Eli Lilly's CETP inhibitor evacetrapib showed significant promise in early Phase III results, with the potential to reduce major cardiac events.⁴⁷ However, other CETP inhibitors from Pfizer and Roche struggled with safety in early trials and stalled in development. Specifically, Pfizer's torcetrapib was halted in 2006 after excessive deaths in Phase III trials, and Roche's dalcetrapib was halted due to lack of clinically meaningful efficacy. These results led to significant concerns over evacetrapib and the rest of the CETP inhibitor class.

Furthermore, there were existing treatments available for FH patients who might not want to alter their treatment mechanisms. While several cholesterol-lowering drugs could help with HeFH, as of 2013 there were two approved treatments for HoFH: Juxtapid, manufactured by Aegerion Pharmaceuticals, and Mipomersen, manufactured by Genzyme Corporation. These drugs were priced at approximately \$300,000 per year and \$176,000 per year, respectively. Juxtapid took the form of a capsule while Mipomersen was an injectable. Though helpful in lowering LDL levels, these treatments often still did not lead to HoFH patients reaching their targets. Similarly, existing treatments for HeFH patients helped lower LDL, but would also often not lead to patients reaching their cholesterol targets.

Finally, Merck's existing non-statin treatment, Zetia, had a potential advantage over the PCSK9 inhibitors, as it was not an injectable but instead a small molecule. Zetia reduced LDL levels by inhibiting the absorption of cholesterol from the small intestine. Though its efficacy was not nearly as strong as that of PCSK9 inhibitors, Zetia was still found to lower LDL levels by 18% and could appeal as a lower-cost option to insurance companies and non-injectable option to patients.⁴⁸ In addition, the combination therapy of Zetia with Zocor, called Vytorin, was shown to reduce serious cardiac events in patients who had previously suffered heart attacks or chest pains, as compared to statins alone, dropping from 34.7% to 32.7%, a statistically significant difference.⁴⁹ With positive

results in reducing serious cardiac events and less invasive treatment, it was possible that insurers, physicians, or patients could choose this kind of drug over PCSK9 inhibitors, further increasing the competition in the LDL cholesterol-lowering drug market. Zetia cost about \$3,600 per year but would lose patent protection in 2016.

The Challenge

After Bush left the meeting, Alex Shamoun promptly called his team together. He looked at the group and said, “Congratulations on developing a strong marketing plan. Sheila thought it was terrific. We just need to develop a forecast for 2018 U.S. revenues. Let’s get to work.”

Exhibit 1: Amgen Financial Highlights (\$ in billions)

	2012	2013	2014
Sales	17.3	18.7	20.1
COGS	2.1	2.1	2.1
SG&A	8.1	9.0	8.9
R&D	3.4	4.1	4.3
Operating income	6.1	6.3	6.9
Net income	4.3	5.1	5.2

Source: Amgen Annual Reports.

Exhibit 2: Amgen's Largest Products (\$ in billions)

	2012	2013	2014
Neulasta/Neupogen	5.4	5.8	5.8
Enbrel	4.2	4.6	4.7
Aranesp	2.0	1.9	1.9
Epogen	1.9	2.0	2.0
Xgeva	0.7	1.0	1.2
Prolia	0.5	0.7	1.0
Sensipar/Mimpara	1.0	1.1	1.2

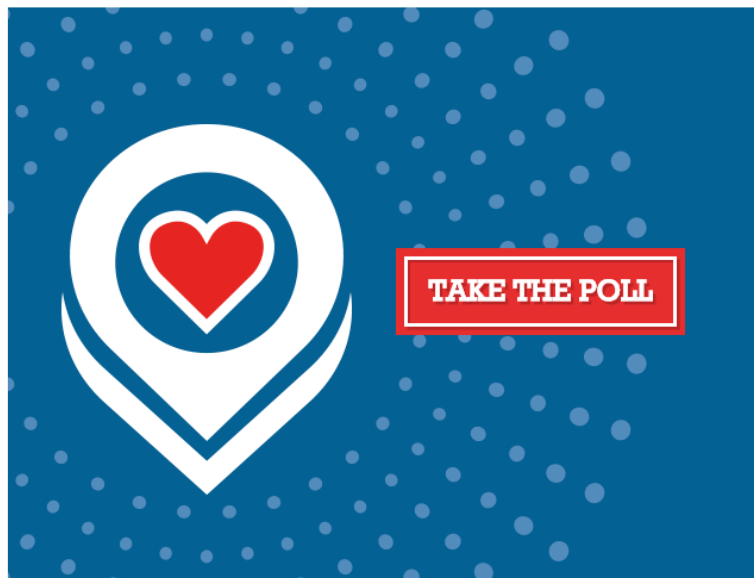
Source: Amgen Annual Reports.

Exhibit 3: Selected Trial Results for PCSK9 Inhibitors

Company/Drug	Clinical Stage	Summary of Results
Amgen: Repatha	Phase II	Lowered LDL by 51% in patients intolerant of statins after 12 weeks
	Phase III	Reduced LDL by 57% over placebo on patients with LDL levels higher than 75 mg/dL
	Phase III	For hypercholesterolemic patients, lowered LDL 55–57% over placebo, and 38–40% over Zetia
	Phase III	Reduced LDL levels by 75% over placebo when combined with a statin
	Phase III	For patients with HoFH, reduced LDL levels by 31% compared with placebo; 40% greater reduction in LDL than Zetia
	Phase III	Reduced LDL levels by 53–56% for patients who failed on four or more statins; 75% of patients achieved LDL levels lower than 100 mg/dL
	Phase III	Reduced rate of major cardiac events from 2.1% if taking just a statin to 0.95% if taking a statin and Repatha
Sanofi/Regeneron: Praluent	Phase III	After 24 weeks, reduced LDL by 54% in patients at high risk of cardiovascular disease over placebo
	Phase III	Reduced LDL levels by 45% in patients intolerant of statins; ten times more patients reached cholesterol goals than those on Zetia in trial
	Phase III	For patients with HeFH, reduced LDL by 46% as opposed to 7% for placebo
	Phase III	Reduced LDL levels by 60% over placebo for patients with high cardiovascular risk
	Phase III	Half the risk of major cardiovascular event for high-risk patients (3% versus 1.4% for placebo)
Pfizer: bococizumab	Phase II	Reduced LDL by 52% on patients also taking statins
	Phase II	Reduced LDL levels by up to 53 mg/dL over placebo at all dosing levels

Exhibit 4: Sanofi/Regeneron’s “Cholesterol Counts” Campaign**WELCOME**

How much do you know about cholesterol? We want to hear from you to help put cholesterol back on the map! With your participation, the *Cholesterol Counts* program will get a snapshot of America's cholesterol knowledge. By answering a few quick questions and encouraging others to do the same, you will be part of the movement to establish America's Cholesterol Count.



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